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4. A satellite orbiting the Earth completes one revolution in 105 minutes.

Mass of the Earth = 6.0×10^{24} kg, radius of the Earth = 6400 km

(a) Explain what is meant by the term *radian*. [2]

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(b) Calculate the satellite's angular velocity, ω, in rad s^{-1}. [3]

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(c) The gravitational force on the satellite is given by $\frac{GM_E m}{R^2}$ where m is the mass of the satellite, M_E is the mass of the Earth and R is the radius of the orbit. Show that:

$$R = \sqrt[3]{\frac{GM_E}{\omega^2}}$$

[2]

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(d) Determine the height of the satellite above the surface of the Earth. [3]

(e) The mass of the Moon is 7.3×10^{22} kg and its radius is 1 740 km. Discuss whether a satellite would be able to orbit the Moon with a period of 105 minutes. [3]

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